

Additional Admitted Tangents for Joint Nine-Point Certificates and Simple Critical Zeros of the Riemann Zeta Function

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Abstract

We strengthen a joint nine-point certificate for a fixed perturbed Montgomery–Taylor window. An expanded capture retains 241 low-reward cells of an admitted eight-point certificate. Their reflections have 2,399 feasible ordered intersections. Existing certified point values, derivative intervals and convex regions supply additional tangent cuts, without a new kernel table. A standard-library rational verifier checks every intersection with complete finite-box residual corrections. The uniform nine-point reward increases to $805260/10^8$. The lossless separation and all-zero spectral transport then give

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)} \geq \frac{66812491}{99194740} = 0.6735487284910470 \dots$$

Here N counts all nontrivial zeros with multiplicity and N_0^s counts simple zeros on the critical line. The proof assumes no RH, zero-free strip or higher correlation conjecture. A companion Lean formalization now proves the fixed nine-point inequality and its dyadic and cumulative simple-zero corollaries. Fresh axiom audits contain only `propext`, `Classical.choice` and `Quot.sound`. This short paper records the certificate and its exact transport, with the analytic framework and original data credited to their prior sources.

1 Statement and provenance

Let $N(T)$ count the nontrivial zeros of ζ with imaginary part in $(0, T]$, including multiplicity. Let $N_0^s(T)$ count those zeros that are simple and have real part $1/2$.

Theorem 1.1. *The simple critical-line zero count satisfies*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)} \geq \frac{66812491}{99194740}.$$

The complete exact certificate of Section 4 for the fixed window, point data and convexity regions, and the proved analytic transport establish this bound without an additional unproved analytic hypothesis. The companion formalization discharges the continuous and finite certificate inputs.

The original window and eight-point weights come from the fixed public AM submission [1]. The separation mechanism is adapted from Knausgård [2, Section 5]. Our preceding nine-point paper [3] obtained $66812491/99194897$. The present improvement uses more of the already admitted point data and recaptures a wider low set before proving a larger uniform reward. The exact gains are

$$\begin{aligned} \frac{66812491}{99194740} - \frac{66812491}{99194897} &= \frac{10489561087}{9839612017241780}, \\ \frac{66812491}{99194740} - \frac{1669159}{2478195} &= \frac{517622017}{49164781738860}. \end{aligned}$$

These are respectively about 0.0001066054 and 0.0010528309 percentage points. The second is a comparison with a fixed arXiv v1, not an exhaustive global priority claim. Neither zero-free boundaries nor the original arithmetic fourth-moment estimates are improved here.

2 Fixed window and local data

Put $\beta = 1/\sqrt{2}$, $Z = \sqrt{2} \sin \beta$ and $I = [-1/2, 1/2]$. On I , with zero extension outside it, set

$$f(u) = \frac{\cos(\sqrt{2}u) + \sum_{j=1}^{12} c_j \cos(2\pi j u)}{Z}, \quad K(t) = \int_I f(u) e^{-2\pi i t u} du, \quad w(t) = K(t)^2.$$

The numerators $10^9 c_j$ are

$$(12310798, -15041681, 3867664, 6489926, -992327, 5580097, \\ -6846472, 3781297, -5670353, 3355089, -450523, -218483).$$

The density is positive and has integral one, so $0 \leq w \leq 1$ on the real axis. Define

$$F_8(h) = \sum_{r=0}^6 b_r h_r + \sum_{0 \leq i < j \leq 7} a_{ij} w(h_i + \dots + h_{j-1}),$$

where

$$10^8 (b_r)_{r=0}^6 = (28898, 57272, 75526, 80958, 75526, 57272, 28898), \quad B = \sum b_r = 404350/10^8.$$

For each row $i = 0, \dots, 6$, the entries $10^8 a_{i,i+1}, \dots, 10^8 a_{i,7}$ are

$$(13630830, 0, 19565904, 20317441, 45030671, 100000000, 200000000), \\ (29997996, 30621878, 47766489, 79682558, 109938656, 100000000), \\ (37356140, 69378121, 65335210, 79682558, 45030671), \\ (38030065, 69378121, 47766489, 20317441), \\ (37356140, 30621878, 19565904), \quad (29997996, 0), \quad (13630830).$$

There are 26 nonzero terms. For every index span $s = 1, \dots, 7$, $\sum_i a_{i,i+s} \leq 2$.

Computational input 2.1. *The fixed continuous PC8 certificate gives $F_8(h) \geq c_0$ on $[0, \infty)^7$, where $c_0 = 805003/10^8$. Its point lists certify values and derivative intervals of w ; its packed REG data certify convexity on their real intervals. The corresponding minorant, empty-cell and root-cursor checks have the continuous semantics described in [3, 7].*

We reuse this disclosed input. The original 240 compiled integer checks are not rerun or relabelled as 240 kernel proofs. The present capture reruns the 41 required outer/root checks; additional rational checks use the same admitted point and region data.

3 Additional interval-valid tangents

Let $SC = 32768$ and $E = 10^9$. A certified point p with pack

$$V + 2^{32}P + 2^{64}M$$

provides, for $q = p/SC$,

$$v = V/10^{10} \leq w(q), \quad d_- = (P - 2E)/E \leq w'(q) \leq d_+ = (2E - M)/E.$$

Point values and derivatives alone do not establish an interval tangent. The following direct branch of the original convex-region test does.

Lemma 3.1. *For a leaf term with actual integer interval $L \leq U$, put $L^* = \min(L, p)$, $U^* = \max(U, p)$ and $k = (p + 16384) \gg 15$. Let A_k, B_k be its admitted **REG** endpoints. If*

$$k < 16, \quad A_k \leq L^* \leq p \leq U^* \leq B_k, \quad P, M \geq 1,$$

then $w(x) \geq v + w'(q)(x - q)$ for every $x \in [L^/SC, U^*/SC]$. In particular it holds on the actual leaf interval.*

Proof. The admitted region makes w convex on $[A_k/SC, B_k/SC]$. The supporting tangent at q lies below w there, and replacing $w(q)$ by its lower bound v preserves the inequality. In the original **tcheckPZ**, all containment and positive derivative-pack tests pass, and both extension disjunctions pass by containment alone. Thus its $n = 1$ branch also gives the same full-interval assertion. The case $L^* = U^* = p$ passes its singleton branch and is covered as well. Each pack is taken from its original list **PTL_i** with its own **PTF** assertion; no new merged-table assertion is presumed. $\square$

Lemma 3.2. *For an interval-valid point tangent on $[l, u]$ containing q , both functions*

$$v + d_+(l - q) + d_-(x - l), \quad v + d_-(u - q) + d_+(x - u)$$

lie below $w(x)$ on $[l, u]$.

Proof. Write $d = w'(q)$. The tangent minus the first line is $(d - d_-)(x - l) + (d_+ - d)(q - l) \geq 0$; minus the second it is $(d_+ - d)(u - x) + (d - d_-)(u - q) \geq 0$. $\square$

Every admitted extra point is tested separately by Lemma 3.1. All its lines can then be imposed on the same actual square $w(x)$. The old minorant's choice of at most two mixture points is no restriction on this use. For zero-width leaf spans the actual $U = L$ is retained; the enlarged interval used for a constant value lookup is not a tangent domain. We do not take an invalid maximum of the two raw derivative endpoint tangents across q .

4 Expanded cover and exact certificate

For $g \in [4/5, \infty)^8$, set

$$F_9(g) = \frac{1}{2}\{F_8(g_0, \dots, g_6) + F_8(g_1, \dots, g_7)\} + 2w(g_0 + \dots + g_7).$$

We prove the uniform reward $c_* = 805260/10^8$, an increment $c_* - c_0 = 2.57 \cdot 10^{-6}$. The nine-point pair weights are the half-sum of the old frame weights plus two on span eight. All index-span capacities are at most two; the total gap pressure remains B .

Use the larger capture threshold $c_L = 805803/10^8$. If either old frame has $F_8 \geq c_L$, then

$$F_9 \geq (c_L + c_0)/2 = 805403/10^8 > c_*.$$

It remains to cover the two-frame low case. The original clear COV domains pay $c_0 + (4/5)B$, and the large-gap domains pay $c_0 + (4/5)(B - \max b_r)$; both extra fees exceed $c_L - c_0 = 8 \cdot 10^{-6}$. The small-gap domain is absent on $[4/5, \infty)^7$. Hence the original 32 root trees and their bad intervals suffice.

The capture preserves the original reward comparisons, return values, validity guards and bit cursors. Only the diagnostic clone of the final reward comparison changes to 805803; its reward search uses fuel ten. All 41 outer/root checks pass. A leaf whose stronger comparison passes excludes the low point; a successful old leaf whose stronger comparison fails is recorded. The initial capture handles the half-space $h_6 \geq h_0$. Direction leaves with $u_6 < l_0$ cannot contain a point in that half-space; empty-cell leaves contain no feasible point. This yields 241 original cells.

The same band-limited majorant is

$$g(u) = \frac{61}{100} \{ \text{sinc}(\frac{4}{5}(u - 1/2)) + \text{sinc}(\frac{4}{5}(u + 1/2)) \}^2.$$

The continuous certificate of [4, 3] proves $g \geq f$, $\hat{g}(t) = 0$ for $|t| \geq 4/5$, and $\int g \leq 61/32 < 2$. With fixed smoothing $f_\epsilon = f\chi_\epsilon^2/m_\epsilon$, $m_\epsilon \rightarrow 1$, let $d_\epsilon = \|f_\epsilon - f\|_1 \rightarrow 0$. For $m_\epsilon > 61/64$ the Gram matrix of the separated set has norm below two, so the spectral clipping at level two causes no loss. The total local pair weight is less than sixteen; each squared overlap changes by at most $2d_\epsilon$. The sliding error is therefore at most $32d_\epsilon m$.

Take a maximal disjoint collection of pairs at distance less than $4/5$ among the n simple critical-line points. The remaining set is separated. For each close pair the same window satisfies $K(t) > 134/825$, hence $K(t)^2 > 1/40 > c_*$. Its two-by-two Gram spectrum lies in $[0, 2]$, and its reward pays $2c_*$ with smoothing error at most $4d_\epsilon$. Convex spectral-trace pinching combines the pairs and the separated set. For $\varphi_2(t) = t^2 - (t - 2)_+^2$ and their full Gram matrix U ,

$$J(U) := \text{tr } \varphi_2(U) - n \geq c_* n - B \text{span}(y) - 8c_* - 32d_\epsilon n.$$

This includes empty or short separated sets and every close pair.

The all-zero Hermitian operator A retains all off-line reflection pairs and multiplicities. Its remainder Q after the n simple critical-line columns has positive inertia p with $N - n \geq 2p$. The threshold trace lemma of [3, 2] gives

$$\text{tr } A^2 \geq 2(N - n) + \text{tr } \varphi_2(U) = 2N - n + J(U).$$

The unconditional pair-correlation formula with fixed smoothing gives $\text{tr } A^2 = (C(f_\epsilon) + o(1))N$ [5, 6]. The removal of its correlation weight and the order of limits are the same as in [3]: first $T \rightarrow \infty$ for fixed ϵ , then $\epsilon \rightarrow 0$. The full height span is at most $N + o(N)$. The bounded $8c_*$ edge term disappears, and the smoothing error tends to zero. Consequently

$$\liminf \frac{n}{N} \geq \frac{2 - C(f) - B}{1 - c_*}.$$

The same window-energy certificate gives

$$C(f) = \frac{1}{2} + \beta \cot \beta + \frac{\sum_j c_j^2 (1/2 - 1/(4\pi^2 j^2))}{2 \sin^2 \beta}, \quad 2 - C(f) > 67216841/10^8.$$

Thus the last lower bound exceeds

$$\frac{67216841 - 404350}{10^8 - 805260} = \frac{66812491}{99194740},$$

proving Theorem 1.1. A fixed-width $7/8$ strip is not used to turn off-line zeros into real Gram points or to reduce their inertia budget.

6 Reproduction and trust boundary

The committed data are `output/am-expanded-nine-point-certificate.json`; the standard library checker is `scripts/am_expanded_nine_point_certificate.py`. From the repository root run

```
python scripts/am_expanded_nine_point_certificate.py --check
python scripts/am_expanded_nine_point_certificate.py --replay
```

The first command checks stored integers and all rational duals. The second additionally compares the used packs and convex-region endpoints with the fixed upstream source, regenerates an isolated Lean capture with Lean 4.34.1, and compares every emitted field. The source must be present at the path documented by the existing PC8 replay. Neither command imports an upstream theorem as a new axiom.

The original computational presentation used the disclosed PC8 minorant, point-table and convexity admission, together with the complete capture semantics. Integer runtime replay and standard-library rational checks are distinct from Lean kernel proof checking. Independent full-domain reconstruction and the build identities remain recorded in the repository. The headline native-computation axiom of [2] is not consumed.

The separate formalization repository uses Lean 4.33.0-rc2 and the pinned Mathlib and Zeta23 versions of the website contract. Its continuous nine-point certificate closes all point, geometry and finite-integer premises. The terminal theorems `simple_dyadic` and `simple_cumulative` belong to the namespace `RHWeilRecord.SimpleCorollary`. They use Zeta23’s actual simple and total zero-count definitions, with no remaining numerical or analytic premises. Their full module build and fresh axiom audits pass; the only axioms are `propext`, `Classical.choice` and `Quot.sound`. Reproducible records are provided in `verification/` of that repository.

The three website entry declarations use the distinct critical-line count, obtained from the simple count by its proved inclusion. All three compile from the complete module chain and pass independent nanoda checking. Single-file packaging, the website’s resource limits and official acceptance remain separate obligations; no website record is claimed here. Mathematical research remains paused during submission engineering.

References

- [1] *AM eight-point PC8CL certificate*, fixed public submission `dani-bound-2026`, commit `d272437e7eb17b87c8c3d7cece592aa23785ab`, proof source.
- [2] K. M. Knausgård, *More than 83.9% of the zeros of the Riemann zeta function are distinct and more than 67.35% are simple and on the critical line*, arXiv:2610.08965v1, October 6, 2026, Section 5.
- [3] RH–Weil research project, *Joint Minorants for Nine Points and Simple Critical Zeros of the Riemann Zeta Function*, October 8, 2026, fixed manuscript.
- [4] RH–Weil research project, *Lossless Assembly of an Eight-Point Certificate for Simple Critical Zeros of the Riemann Zeta Function*, October 8, 2026, fixed manuscript.
- [5] S. A. C. Baluyot, D. A. Goldston, A. I. Suriajaya and C. L. Turnage-Butterbaugh, *An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta function*, Acta Arith. 214 (2024), 357–376, arXiv:2306.04799; see also the correction in arXiv:2501.14545v3, Section 3.
- [6] Y. Lamzouri, *A new proof that more than 2/3 of the zeros of the Riemann zeta function are simple and on the critical line*, arXiv:2609.02882v1, Section 3.
- [7] RH–Weil research project, public repository, notes 483, 501–502; PC8 continuous audits and finite replay; expanded nine-point certificate and independent reconstruction.